Hybrid Unsupervised Risk Phenotyping and Supervised Classification for Breast Cancer Diagnosis

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CS439 Final Project
Code repository: `https://github.com/daniyalfs/cs439-final-project.git`

Abstract

Early identification of malignant breast masses is a high-impact classification problem because diagnostic decisions require both predictive accuracy and interpretability. This project studies whether unsupervised morphology-based phenotypes can support binary diagnosis prediction on the Wisconsin Diagnostic Breast Cancer dataset. We compare a majority-class dummy model, logistic regression, random forest, and radial basis function support vector machine against a proposed hybrid pipeline that first learns K-Means phenotypes from standardized cell-nuclei measurements and then appends one-hot cluster indicators to a logistic regression classifier. All preprocessing, scaling, clustering, and encoding steps are fit only on the training set to reduce leakage. On a stratified 20% held-out test set, the best accuracy is achieved by the RBF SVM at 98.25%, while the random forest obtains the strongest ROC-AUC at 0.9970. The hybrid K-Means + Logistic model achieves 96.49% accuracy and 0.9927 ROC-AUC. The hybrid model does not outperform the strongest supervised baselines, but it provides a transparent phenotype layer that helps explain the structure of the feature space and supports clinically readable error analysis. These results suggest that unsupervised phenotyping can be useful for interpretability, but the WDBC dataset is already highly separable using standard supervised models.

1 Introduction

Breast cancer diagnosis often depends on distinguishing benign from malignant tissue using measurements derived from medical imaging or biopsy procedures. From a data science perspective, this is a binary classification problem where false negatives are especially costly because a malignant case predicted as benign can delay treatment. A useful model should therefore achieve strong recall for malignant cases while remaining interpretable enough for human review.

This project asks the following research question: *Can unsupervised morphology-based risk phenotypes learned with K-Means improve or explain supervised prediction of malignant versus benign breast masses?* The project uses the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, which contains 569 samples and 30 real-valued features computed from digitized fine needle aspirate images of breast masses [1]. These features describe nucleus size, shape, texture, concavity, and related morphology measurements.

The main contributions are threefold. First, we build a leakage-aware preprocessing pipeline that remaps the binary diagnosis target, standardizes numeric predictors, learns K-Means clusters only on training data, and one-hot encodes cluster membership before supervised training. Second, we compare the proposed hybrid model against multiple baselines using accuracy, precision, recall, F1, ROC-AUC, average precision, and confusion-matrix error analysis. Third, we provide PCA visualizations and permutation feature importance to connect predictive results with interpretable morphology patterns.

2 Related Work

Breast cancer diagnosis with machine learning. The WDBC dataset is widely used as a benchmark for supervised diagnosis because its features are quantitative measurements extracted from breast-mass cell nuclei [1]. Prior work has applied supervised classifiers such as multilayer perceptrons, support vector machines, random forests, and feature-selection pipelines to this dataset, often reporting high performance because malignant and benign cases are strongly separable in several morphology dimensions [7].

Supervised classification baselines. Logistic regression is a strong linear baseline for binary classification because it estimates class probabilities through a regularized logit model. Random forests combine many decision trees trained with randomized feature and sample variation, reducing variance and capturing nonlinear interactions [3]. Support vector machines

maximize a separating margin and, with a radial basis function kernel, can model nonlinear class boundaries [4]. These models are appropriate baselines because they represent increasing levels of complexity: linear, tree ensemble, and nonlinear margin-based classification.

Unsupervised phenotyping and visualization. K-Means clusters data by minimizing within-cluster squared distances around k centroids [5]. In this project, K-Means is not used as the final classifier. Instead, it creates a morphology phenotype variable that is one-hot encoded and added to the supervised model. PCA is used only for visualization by projecting standardized 30-dimensional data into two principal components that explain the largest variance directions [6]. The implementation uses scikit-learn, a standard Python machine learning library [2].

3 Methodology

3.1 Dataset and Target Definition

The dataset contains 569 patient-level samples, 30 numeric predictors, and a binary diagnosis label. The original scikit-learn target uses 0 for malignant and 1 for benign, so we remap it to a more natural modeling target:

$$y_i = \begin{cases} 1, & \text{if diagnosis is malignant,} \\ 0, & \text{if diagnosis is benign.} \end{cases}$$

This makes malignant diagnosis the positive class for precision, recall, F1, ROC-AUC, and average precision. The class distribution is 212 malignant samples and 357 benign samples. We use an 80/20 stratified train/test split with random seed 42, producing 455 training samples and 114 held-out test samples.

3.2 Preprocessing and Leakage Prevention

The preprocessing design follows a strict train/test separation. The split is performed before scaling, clustering, encoding, or model fitting. The main numeric features are standardized with $z = (x - \mu)/\sigma$, where μ and σ are learned from the training set only and then applied to the test set. This is important because K-Means and regularized logistic regression are scale-sensitive.

The raw dataset does not contain source categorical predictors. To satisfy the project requirement for categorical processing in a meaningful way, We create a derived categorical variable using K-Means cluster membership. The number of clusters is selected using training-only silhouette score over $k \in \{2, 3, 4, 5, 6\}$. The selected model uses $k = 2$. The learned cluster labels are categorical phenotype IDs, so they are transformed using one-hot encoding. If $c_i \in \{1, \ldots, k\}$ is the cluster assignment for sample i, the encoder produces a vector $e(c_i)$ whose entries indicate cluster membership. The final hybrid feature vector is

$$z_i^{hybrid} = [\text{StandardScaler}(x_i), e(c_i)].$$

The one-hot encoder is fit on training cluster labels only. Test samples receive cluster labels through the training-fitted K-Means model and are then encoded through the training-fitted encoder.

3.3 Models

We compare five models:

1. **Majority Dummy**: predicts the most frequent class and acts as a lower-bound baseline.
2. **Logistic Regression**: standardized predictors with class-balanced logistic regression.
3. **Random Forest**: 100-tree class-balanced random forest.
4. **RBF SVM**: standardized predictors with a radial basis function support vector machine.
5. **Hybrid K-Means + Logistic**: standardized predictors plus one-hot encoded K-Means phenotype indicators, followed by class-balanced logistic regression.

The hybrid model is the proposed pipeline. It tests whether a compact unsupervised representation of morphology groups adds useful predictive or interpretive information beyond raw standardized measurements.

3.4 Evaluation Metrics

The primary metrics are accuracy, precision, recall, F1, ROC-AUC, and average precision. Recall is emphasized because malignant false negatives are the most concerning error type. ROC-AUC evaluates ranking quality over all thresholds, while average precision is useful because the positive class is smaller than the benign class. We also report confusion matrices and permutation importance. Permutation importance measures how much ROC-AUC decreases when a feature is randomly shuffled, which helps identify influential morphology measurements.

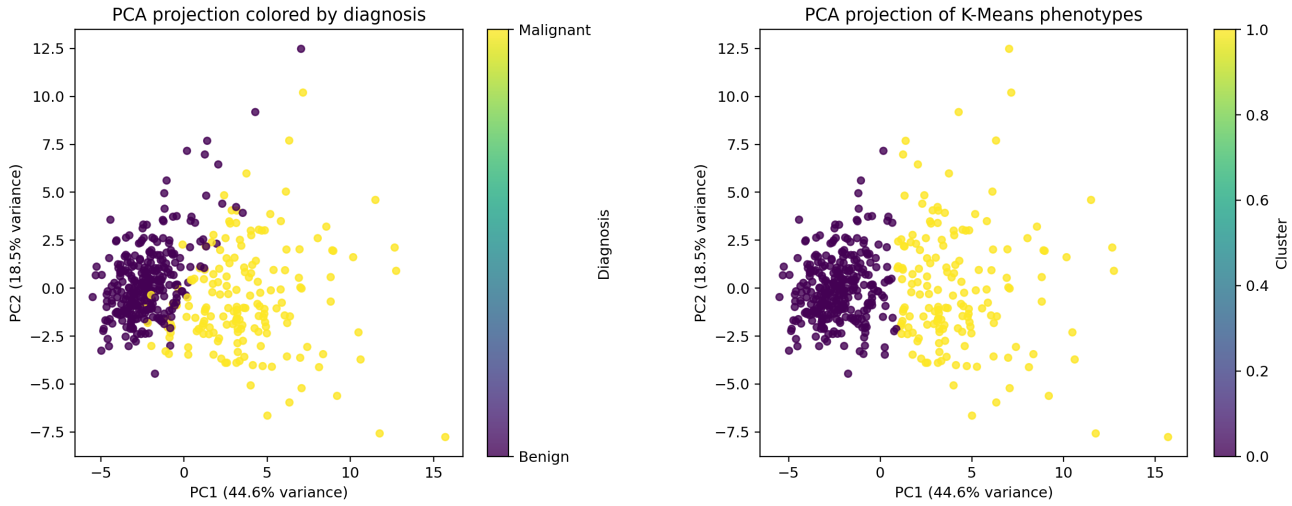
4 Experiments and Results

4.1 K-Means Model Selection

K-Means was selected using silhouette score on the training data only. Table 1 shows that $k = 2$ produced the highest silhouette score, which is also intuitive for a dataset with two diagnostic outcomes. This does not mean that the clusters are equivalent to diagnosis labels; rather, it suggests that a two-phenotype morphology partition is the most stable among the tested values.

Table 1: Training-only K-Means model selection.

k	Silhouette score	Inertia
2	0.3432	9254.8273
3	0.3233	8075.3173
4	0.2782	7344.3357
5	0.1565	6810.1568
6	0.1604	6328.7294



(a) PCA colored by diagnosis.

(b) PCA colored by K-Means phenotype.

Figure 1: PCA projections show that the data contains a strong morphology gradient. The K-Means phenotypes broadly follow the same geometry as the diagnostic separation, which supports their use as an interpretable derived categorical feature.

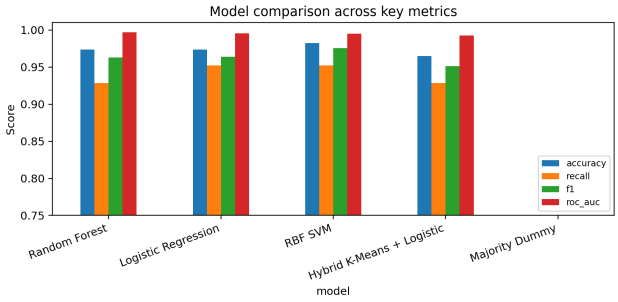
4.2 Held-Out Test Performance

Table 2 reports performance on the 114-sample held-out test set. The RBF SVM achieves the highest accuracy, 98.25%, with only two malignant false negatives and no false positives. The random forest achieves the highest ROC-AUC, 0.9970, and perfect precision at the default threshold, but it misses three malignant cases. Logistic regression is also strong, with 97.37% accuracy and 0.9954 ROC-AUC.

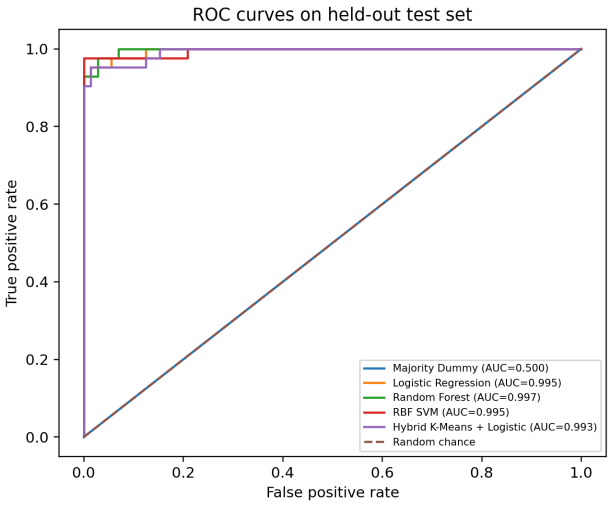
The proposed hybrid model achieves 96.49% accuracy, 92.86% recall, 95.12% F1, and 0.9927 ROC-AUC. It is competitive, but it does not outperform the strongest supervised models. This result is useful because it shows that the K-Means phenotype variable preserves much of the diagnostic signal but does not add enough extra information to beat the direct nonlinear baselines.

Table 2: Held-out test metrics. Positive class is malignant.

Model	Acc.	Prec.	Recall	F1	ROC-AUC	Avg. Prec.	TN	FP	FN	TP
Random Forest	0.9737	1.0000	0.9286	0.9630	0.9970	0.9952	72	0	3	39
Logistic Regression	0.9737	0.9756	0.9524	0.9639	0.9954	0.9931	71	1	2	40
RBF SVM	0.9825	1.0000	0.9524	0.9756	0.9950	0.9937	72	0	2	40
Hybrid K-Means + Logistic	0.9649	0.9750	0.9286	0.9512	0.9927	0.9896	71	1	3	39
Majority Dummy	0.6316	0.0000	0.0000	0.0000	0.5000	0.3684	72	0	42	0



(a) Key metric comparison.



(b) ROC curves.

Figure 2: Supervised baselines and the hybrid model all strongly outperform the dummy baseline. The hybrid model remains close to the strongest baselines but is slightly weaker on recall and ROC-AUC.

4.3 Cross-Validation and Robustness

To check whether the held-out split was unusual, We also ran 3-fold stratified cross-validation for the main supervised baselines. Logistic regression and RBF SVM both achieved mean accuracy of 0.9754. RBF SVM had the highest mean ROC-AUC at 0.9951, while logistic regression had 0.9939. Random forest remained strong but had higher recall variation across folds. This supports the held-out result that the dataset is highly separable and that simpler standardized baselines are already competitive.

Table 3: Three-fold stratified cross-validation summary for supervised baselines. Values are mean $\pm$ standard deviation.

Model	Accuracy	Recall	F1	ROC-AUC
Logistic Regression	0.9754 ± 0.0121	0.9624 ± 0.0406	0.9668 ± 0.0164	0.9939 ± 0.0050
Random Forest	0.9526 ± 0.0320	0.9154 ± 0.0975	0.9332 ± 0.0489	0.9907 ± 0.0104
RBF SVM	0.9754 ± 0.0080	0.9671 ± 0.0325	0.9669 ± 0.0112	0.9951 ± 0.0053

4.4 Feature Importance and Error Analysis

Permutation importance was computed for the strongest non-hybrid model by held-out ROC-AUC, which was the random forest. The most important features are worst concave points, worst area, worst perimeter, worst radius, and worst concavity. These variables are clinically intuitive because they describe extreme values of cell-nucleus shape and size, which are relevant to malignancy. The confusion matrices show that all strong models make very few errors, but the remaining errors are mainly malignant false negatives. This reinforces the importance of threshold analysis in any real clinical setting. A model with slightly lower overall accuracy may still be preferred if its threshold can be adjusted to reduce false negatives.

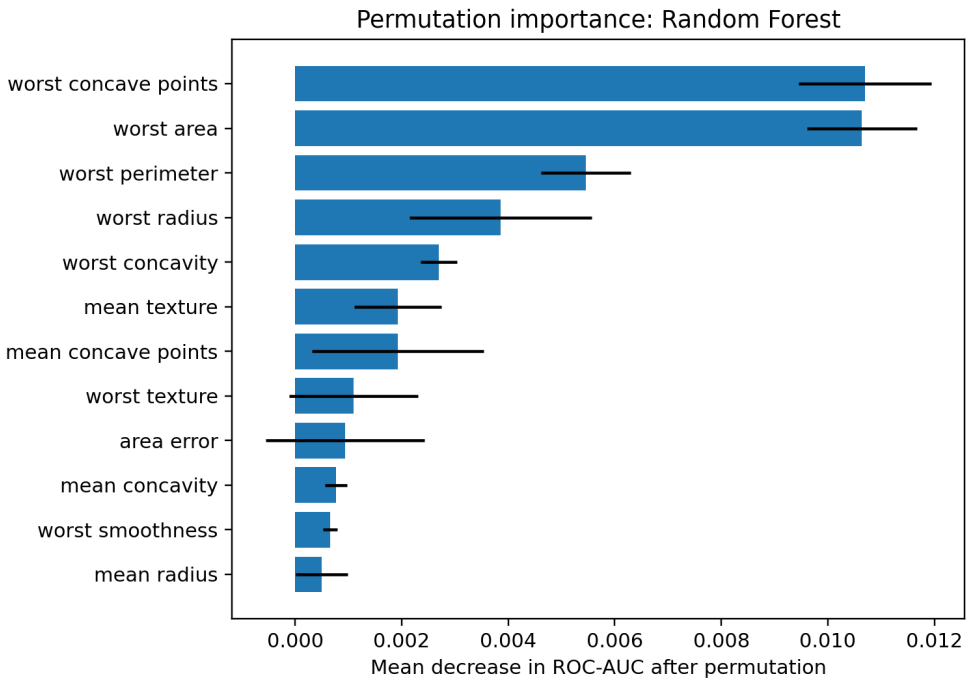


Figure 3: Permutation importance for the best ROC-AUC baseline. The strongest signals are worst-case concavity, area, perimeter, and radius measurements.

5 Conclusion

This project developed and evaluated a hybrid unsupervised-supervised diagnosis pipeline for the WDBC dataset. The strongest held-out accuracy came from the RBF SVM at 98.25%, and the strongest ROC-AUC came from random forest at 0.9970. The proposed hybrid K-Means + Logistic model achieved 96.49% accuracy and 0.9927 ROC-AUC. Therefore, the main finding is not that unsupervised phenotyping improves predictive performance, but that it provides a compact and interpretable representation of morphology structure while preserving most of the discriminative signal.

The project has several limitations. The dataset is small, clean, and already highly separable, so performance may overestimate real-world generalization. The features are precomputed rather than learned directly from medical images. The K-Means step assumes spherical clusters in standardized Euclidean space, which may be too simple for complex biomedical subtypes. Future work should test the pipeline on larger multi-site datasets, evaluate calibration and threshold tuning, compare alternative clustering methods such as Gaussian mixture models or spectral clustering, and validate whether phenotype groups correspond to medically meaningful subtypes.

References

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